

# PROBABILITY & STATISTICS PROJECT

## Task-01

- Collect the data of **Dependent variable** (assigned to your group) and **Independent Variables** (keeping in mind literature review).
- Collect the data of variables from some authentic Source like WDI, IFS, PBS, SBP, Kaggle, etc.

## Task-02

Discuss Summary Statistics (Mean, Median, Mode, Quartiles) of each variable included in the Model using R/Python output of summary statistics.

## Task-03

- Construct Box and Whisker Plots for all the variables in one diagram with different colors and identify outliers if any.

## Task-04

- Construct Scatter Plot for all the variables on one grid and interpret each part of the grid separately.

## Task-05

- There should be at least five independent variables (at least 50 observations) in the model.
- Run two different models including
  - Multiple Linear Regression Model
  - Machine Learning Model
- In conclusion section compare the predicted values of the models with original values using graphs and MSE.
- Project Contents
  - Introduction
  - Literature Review
  - Description of Variables and Data Description
  - Estimation of the Models (Model Explanation, Equations etc.)
  - Results and Conclusion
  - References

**NOTE:** Submit the following files on Submission Day

- Excel File of Data Collection
- R File of Programming and LaTeX File, **Use only R/Python**
- Pdf File of Final Project

**SUBMISSION DATE: 5<sup>th</sup> May 2026**